

WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY FOR SELF-IMPROVING AGENTS IN ADAPTIVE WORLDS

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ABSTRACT

Self-improving agents often select policy updates using cheap verifiers, but deploying an update can change the environment and the behavior of other agents. We formalize this mismatch through *Improvement Fidelity*, which asks whether proxy improvement preserves the sign and ordering of deployment improvement over the updates an improvement process actually proposes. Our analysis shows that strong global policy evaluation need not imply reliable update evaluation: operator shift and deployment response can create local fidelity errors, while only errors large relative to candidate margins can change the selected update. We introduce PIVOT-KG, a paired differential validator that estimates the correction between proxy and deployment improvement and allocates a fixed high-fidelity evaluation budget by expected reduction in update-selection regret per unit cost. Across controlled response worlds and 90 held-out decision roots, global policy ranking can remain stable as update-level fidelity deteriorates, and proxy and deployment top-1 updates disagree in 51 of 90 roots. Yet response magnitude and ranking flips alone do not predict validation value: PIVOT-KG helps when high-fidelity evidence can resolve uncertainty that matters for the update decision. These results identify Improvement Fidelity and decision-relevant validation as central to reliable self-improvement in adaptive worlds.

1 INTRODUCTION

Self-improvement is usually written as a loop: generate candidates, score them, promote one, and repeat. A self-improving agent ultimately deploys a policy replacement, not an isolated policy value. In a trading system, a larger order changes liquidity; in an auction, a bid changes rivals’ responses; in a recommender, a policy changes future user behavior. A verifier can therefore rank policies correctly while ranking improvements incorrectly. High-fidelity deployment tests can expose the error, but they are often too costly to run on every candidate.

We study a proposed transition $\tau = (\pi, \pi')$. Let V denote the cheap verifier world and let $*$ denote the policy-induced deployment world. Define

$$\Delta_V = J_V(\pi') - J_V(\pi), \quad \Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi).$$

An improvement reversal, $\Delta_V > 0$ but $\Delta_* < 0$, is one salient failure mode. More generally, deployment can change the sign or ordering of candidate improvements, and a fidelity error matters to selection when it changes the selected update or displaces the no-update action. Figure 1 shows the basic sign failure.

We distinguish three response layers. The *direct* layer holds the incumbent world fixed; the *actor* layer lets the environment respond mechanically to the candidate; the *strategic* layer additionally lets other agents adapt. These correspond to progressively stronger deployment responses.

We call the agreement between proxy and deployment improvement over the updates an agent actually proposes *Improvement Fidelity*. It is operator-relative: the relevant population is the set of updates generated by the self-improvement process. Because only some fidelity errors change the update decision, limited high-fidelity evaluation should be targeted to uncertainty that matters for selection.

Our contributions are:

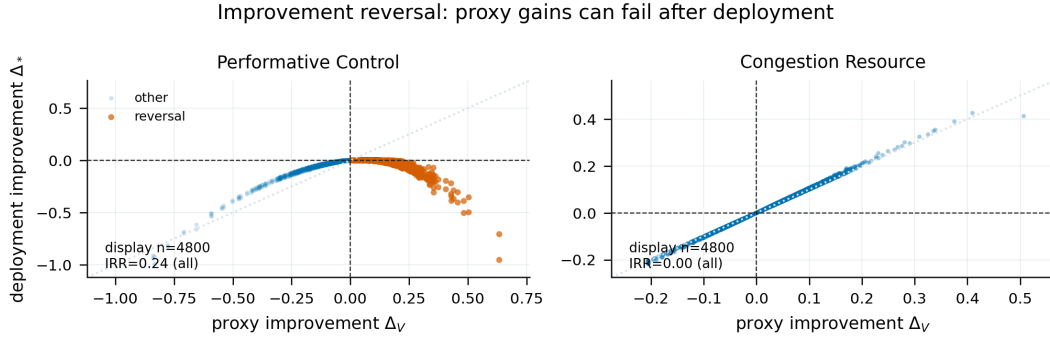


Figure 1: Paired proxy and deployment improvements in the two controlled worlds. The lower-right quadrant is an improvement reversal. Points are a deterministic display subsample; IRR is computed on all transition rows.

Contribution 1. We formalize operator-relative Improvement Fidelity for self-improving agents and show why accurate global policy evaluation can still fail on the updates an agent actually proposes. Our analysis links these failures to operator shift, deployment response, and candidate margins.

Contribution 2. We introduce PIVOT-KG, a paired differential validator that allocates a limited high-fidelity evaluation budget according to the expected reduction in update-selection regret per unit cost.

Contribution 3. We evaluate Improvement Fidelity across controlled direct, actor, and strategic response worlds and held-out decision roots, showing when response creates update-level fidelity errors, when those errors change the selected update, and when targeted high-fidelity validation reduces selection regret.

Together, these contributions provide a direct way to diagnose when self-improvement survives deployment and to spend scarce validation effort where it can affect the update decision.

2 RELATED WORK

Performative and strategic response. Performative prediction and reinforcement learning make deployment part of the data-generating process Perdomo et al. (2020); Mandal et al. (2023); Rank et al. (2024); Mandal & Radanovic (2025); Góis et al. (2025); Piliouras & Yu (2022). These works study learning, stability, and equilibrium behavior under deployment-induced response; we ask whether a particular proposed update preserves its sign and ordering after that response is realized.

World models and policy evaluation. World models and policy-aware simulators evaluate values, rankings, or trajectory fidelity Quevedo et al. (2026); Dann et al. (2026); Hansen et al. (2024). Global value or ranking fidelity can coexist with a wrong local update sign, which is the gap measured here.

Self-improvement and verification. Autonomous policy evolution and verification make iterative self-improvement operational Wang et al. (2026); Zhang et al. (2025); Guo et al. (2026); Wu et al. (2025). We focus on a complementary failure mode: a fixed external verifier can evaluate the wrong response world, so a proposed replacement need not remain an improvement after deployment.

Adaptive selection and paired evaluation. Active learning, knowledge gradients, and best-arm identification allocate expensive observations Settles (2009); Lindley (1956); Frazier et al. (2009); Chick & Inoue (2001); Kalyanakrishnan et al. (2012). PIVOT changes the loss from global prediction error to update-selection regret and uses paired rollouts to cancel common-mode variation.

3 PROBLEM FORMULATION

Policy-induced worlds and response levels. Let $\mathcal{M}[\pi]$ denote the deployment world induced by policy π , including any environment response to that policy, and write $\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi])$. We

consider three response levels:

$$\begin{aligned}\Delta_{\text{direct}} &= J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \\ \Delta_{\text{actor}} &= J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \\ \Delta_{\text{strategic}} &= J_i(\pi', R_{-i}(\pi')) - J_i(\pi, R_{-i}(\pi)).\end{aligned}$$

The direct level holds the incumbent world fixed, the actor level allows the environment to respond to the candidate, and the strategic level additionally allows other agents to adapt. Here R_{-i} denotes a general response map for the other agents; best response is one special case.

Improvement Fidelity. Let an improvement operator $\mathcal{A}$ induce a transition law $Q_{\mathcal{A}}$ over proposed updates $\tau = (\pi, \pi')$. For loss L , we define operator-relative Improvement Fidelity as

$$\text{IF}(V, \mathcal{A}; L) = \mathbb{E}_{(\pi, \pi') \sim Q_{\mathcal{A}}} [L(\Delta_V, \Delta_*)]. \quad (1)$$

The relevant population is therefore the updates the agent actually proposes, rather than the policy space in general. Improvement Fidelity is an update-level quantity: accurate policy values or rankings do not by themselves guarantee accurate improvement estimates.

Core metrics. For transition rows, we measure fidelity-error magnitude by $\text{IDE} = n^{-1} \sum_k |\Delta_{V,k} - \Delta_{*,k}|$ and let ISC denote sign agreement after removing ties. A salient sign failure is the improvement reversal rate $\text{IRR} = P(\Delta_* < 0 \mid \Delta_V > 0)$. For a candidate set C , the decision consequence is measured by improvement-selection regret,

$$\text{ISR} = \max_{j \in C} \Delta_{*,j} - \Delta_{*,\hat{j}}, \quad (2)$$

where $\hat{j}$ is the selected candidate. Thus IDE measures error magnitude, ISC and IRR measure sign preservation, and ISR measures the consequence for update selection.

4 PROPERTIES OF IMPROVEMENT FIDELITY

The following results characterize when policy-level evaluation controls update-level fidelity, how operator shift and deployment response can break that connection, and what a high-fidelity validator must resolve.

Proposition 1 (Uniform value fidelity is sufficient). *If $\sup_{\pi} |J_V(\pi) - \mathcal{J}(\pi)| \leq \varepsilon$, then $|\Delta_V - \Delta_*| \leq 2\varepsilon$. The sign is preserved whenever $|\Delta_*| > 2\varepsilon$.*

Proof. $\Delta_V - \Delta_* = [J_V(\pi') - \mathcal{J}(\pi')] - [J_V(\pi) - \mathcal{J}(\pi)]$; the triangle inequality gives the bound and the sign statement. $\square$

Proposition 2 (Global policy accuracy need not imply update fidelity). *For every $\varepsilon > 0$, there are a finite policy family, a verifier, a policy distribution, and an update operator with policy MAE and rank error below ε , but $\text{IRR} = 1$ and $\text{ISC} = 0$ on the operator’s transition law.*

Proof. Take policies $\pi_0, \dots, \pi_{n-1}$ with $\mathcal{J}(\pi_k) = k/(n-1)$ and swap one adjacent pair in V . Then $\text{MAE} = 2/[n(n-1)]$ and $1 - \text{Spearman} = 12/[n(n^2-1)]$, both tending to zero. Let the operator sample only the swapped directed transition and propose the lower-valued true policy. Its proxy sign is positive and its deployment sign is negative. $\square$

Proposition 3 (Operator Shift Bound). *Let P be a calibration law, let $Q_{\mathcal{A}} \ll P$ have density ratio $w = q_{\mathcal{A}}/p$, and let $\ell(\tau) = L(\Delta_V, \Delta_*)$ have finite variance under P . Then*

$$\mathbb{E}_{Q_{\mathcal{A}}}[\ell] \leq \mathbb{E}_P[\ell] + \sqrt{\text{Var}_P(\ell) \chi^2(Q_{\mathcal{A}} \| P)}. \quad (3)$$

Proof. Write $\mathbb{E}_{Q_{\mathcal{A}}}[\ell] - \mathbb{E}_P[\ell] = \mathbb{E}_P[(w-1)(\ell - \mathbb{E}_P[\ell])]$. Cauchy–Schwarz and $\mathbb{E}_P[(w-1)^2] = \chi^2(Q_{\mathcal{A}} \| P)$ give the result. $\square$

Deployment response provides a second source of update-level error. From the definitions in the previous section,

$$\Delta_{\text{actor}} - \Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi'; \mathcal{M}[\pi]).$$

Thus the actor-direct gap is exactly the change in the candidate’s value caused by the world it induces. The strategic level extends this response to adaptive opponents through R_{-i} .

Lemma 1 (Decision preservation). *For candidates j, k with true margin $m = \Delta_{*,j} - \Delta_{*,k} > 0$, if each estimated improvement has absolute error below $m/2$, then the estimated ordering of j and k is unchanged.*

Proof. The perturbation of the estimated margin has absolute value below m , so the estimated margin remains positive. $\square$

Lemma 2 (Paired variance). *For a paired rollout difference $Y' - Y$,*

$$\text{Var}(Y' - Y) = \text{Var}(Y') + \text{Var}(Y) - 2 \text{Cov}(Y', Y).$$

When shared randomness induces positive covariance, paired evaluation has lower variance than independent evaluation with the same marginal variances.

Together, these results narrow the validation problem. Uniformly accurate policy values control update error, but small global MAE or rank error need not protect the transitions generated by the improvement operator. Operator shift and deployment response can therefore create local fidelity errors, while only errors large relative to candidate margins can change the selected update. Paired high-fidelity rollouts reduce the variance of estimating these differences when common randomness induces positive covariance. This motivates PIVOT-KG: allocate a limited high-fidelity budget to update uncertainty that can affect selection.

5 PIVOT-KG

The preceding results suggest a simple validation principle: spend expensive high-fidelity evaluations on update uncertainty that can change the selection decision. We implement this principle with PIVOT-KG. Given an incumbent and K candidate updates, the cheap verifier first provides proxy improvements $\{\Delta_{V,j}\}_{j=1}^K$.

Rather than model deployment improvement from scratch, PIVOT-KG models the differential correction

$$G_j = \Delta_{*,j} - \Delta_{V,j}.$$

Writing $G = (G_1, \dots, G_K)$, $\Delta_V = (\Delta_{V,1}, \dots, \Delta_{V,K})$, and $\Delta_* = (\Delta_{*,1}, \dots, \Delta_{*,K})$, we place a joint Gaussian prior over the correction vector,

$$G \sim \mathcal{N}(\mu, \Sigma_G), \quad \Delta_* \mid \Delta_V \sim \mathcal{N}(\Delta_V + \mu, \Sigma_G).$$

The prior mean, covariance, and observation variances are estimated from a disjoint calibration set of paired proxy/high-fidelity transitions before held-out evaluation. Thus a new held-out decision root starts from the calibrated posterior $\Delta_V + \mu$ without using its high-fidelity outcomes.

For a queried candidate j , let Y_j denote its paired high-fidelity improvement observation,

$$Y_j = \Delta_{*,j} + \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, s_j^2)$$

where s_j^2 is estimated from calibration data. The incumbent and candidate are evaluated under the same initial state, exogenous randomness, and opponent initialization. Each query updates the joint posterior by Gaussian conditioning; with non-diagonal Σ_G , evidence can also update correlated unqueried candidates.

We augment the candidate set with a no-update action, $C^+ = \{0, 1, \dots, K\}$, where $\Delta_{*,0} = 0$ denotes retaining the incumbent. Let D denote the paired high-fidelity evidence collected at the current decision root. The current decision is

$$a(D) = \arg \max_{j \in C^+} \mathbb{E}[\Delta_{*,j} \mid D].$$

Its posterior expected simple regret is

$$\mathcal{R}(D) = \mathbb{E} \left[\max_{j \in C^+} \Delta_{*,j} - \Delta_{*,a(D)} \mid D \right].$$

For a hypothetical high-fidelity observation Y_j , define

$$\text{EVSI}_j = \mathcal{R}(D) - \mathbb{E}_{Y_j|D} [\mathcal{R}(D \cup \{Y_j\})], \quad A_j = \frac{\text{EVSI}_j}{c_j}. \quad (4)$$

Thus EVSI_j measures the expected reduction in update-selection regret from querying candidate j , while dividing by c_j prioritizes the largest expected regret reduction per unit HF cost. This one-step expected value of sample information gives the knowledge-gradient acquisition used by PIVOT-KG.

Algorithm 1 PIVOT-KG under a fixed high-fidelity budget

Require: Incumbent π ; candidates $\{\pi'_j\}_{j=1}^K$; calibrated correction prior $p(G)$; query costs $\{c_j\}_{j=1}^K$; HF budget B

Ensure: Selected candidate or incumbent

- 1: Evaluate candidates with the cheap verifier to obtain $\{\Delta_{V,j}\}_{j=1}^K$
 - 2: Initialize $D \leftarrow \emptyset$, queried set $Q \leftarrow \emptyset$, remaining budget $b \leftarrow B$, and the calibrated posterior over Δ_*
 - 3: Include the no-update action $j = 0$ with $\Delta_{*,0} = 0$
 - 4: **while** there exists $j \notin Q$ with $c_j \leq b$ **do**
 - 5: **for** each feasible unqueried candidate j **do**
 - 6: Compute EVSI_j and $A_j \leftarrow \text{EVSI}_j / c_j$
 - 7: **end for**
 - 8: $j_q \leftarrow \arg \max_j A_j$
 - 9: Obtain paired HF observation Y_{j_q} under matched randomness
 - 10: $D \leftarrow D \cup \{Y_{j_q}\}$ and update the posterior by Gaussian conditioning
 - 11: $Q \leftarrow Q \cup \{j_q\}$ and $b \leftarrow b - c_{j_q}$
 - 12: **end while**
 - 13:
 - 14: **return** $a(D) = \arg \max_{j \in C^+} \mathbb{E}[\Delta_{*,j} \mid D]$
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The primary comparison uses this fixed-budget rule; posterior-based early stopping is a secondary variant. Unlike global label acquisition, PIVOT-KG values a high-fidelity query by its expected reduction in update-selection regret.

6 EXPERIMENTS

The evaluation follows the progression from update-level fidelity error, to its consequences for update selection, to the value of targeted high-fidelity validation. We first test when global evaluation fails on proposed updates, then whether those errors alter the selected update, and finally whether PIVOT-KG can use limited high-fidelity evidence to reduce selection regret.

6.1 EXPERIMENTAL SETUP

Controlled response worlds use paired common-random-number rollouts and 30 independent seeds. The seed, trajectory, opponent-seed cluster, or held-out root is the inferential unit; transition rows are retained for audit and are not treated as independent replicates. The sealed study contains 30 held-out roots in each of Leduc, Kuhn, and Melting Pot, with 1,800 sealed decisions per cohort. We use Leduc and Kuhn poker through OpenSpiel Lanctot et al. (2019) and the Melting Pot multi-agent suite Leibo et al. (2021). Primary sealed comparisons allocate a fixed HF query budget and compare PIVOT-KG with Uniform; matched-cost analyses additionally include Proxy Only, Global-VOI, Paired LUCB, and an all-HF reference where applicable. HighwayEnv is evaluated separately as a physically reactive external setting. Reported bootstrap intervals resample the corresponding inferential unit. Additional statistical and reproducibility details appear in Appendix E.

6.2 IMPROVEMENT FIDELITY UNDER OPERATOR SHIFT AND DEPLOYMENT RESPONSE

Two mechanisms can break update-level fidelity. Operator shift changes the transition distribution induced by the improvement process, while deployment response changes the world in which the candidate is evaluated. In the operator-shift study, the high-minus-low shift contrast in operator-relative IDE is 0.2627 with bootstrap interval [0.1684, 0.3763] across 30 seeds and 36000 paired rows (Figure 2). Local update error and reversal rates increase while global policy rank remains comparatively stable, showing that strong global evaluation can mask failure on the updates an operator actually proposes.

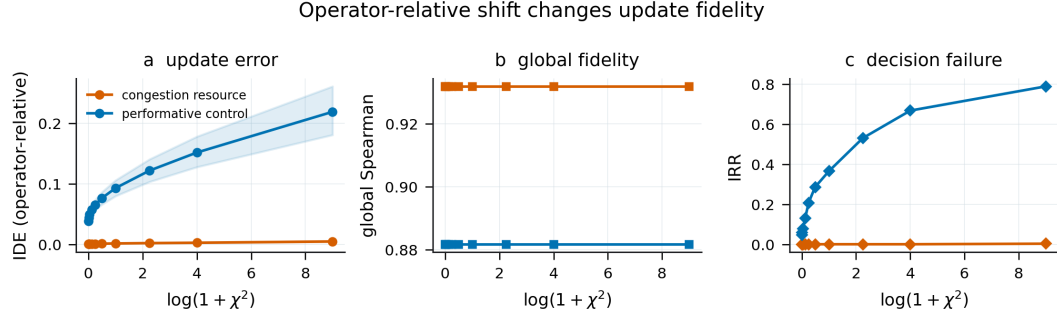


Figure 2: Operator-relative shift diagnostic. Local improvement error rises with registered shift while global policy rank remains comparatively stable. The same environment-by-shift cells show error, rank, and reversal rate.

Deployment response provides a distinct source of error. The direct, actor, and strategic levels progressively expose mechanical environment response and adaptive opponents. In the strategic fixture, the strategic-minus-actor effect is -0.0240 [-0.0249, -0.0231], with a family-balanced strategic reversal rate of 0.9495. Extended world-response decompositions and opponent-family results appear in Appendix A. Together, these experiments identify two distinct sources of update-level fidelity error: operator shift and policy-induced deployment response.

6.3 FROM FIDELITY ERROR TO UPDATE DECISIONS

Lemma 1 connects update error to selection: an error changes an ordering only when it is large relative to the candidate margin. For each sealed root we therefore compute the deployment top-1 margin, the proxy-to-deployment correction, and the largest pairwise correction-to-margin ratio. A ratio at least one marks a proxy/deployment top-1 flip.

Across the 90 sealed roots, 51 have a top-1 flip (18 Leduc, 14 Kuhn, and 19 Melting Pot). To compare across reward scales, Figure 3 reports each root’s squared response gap normalized by the square of its largest candidate range across proxy and long/short deployment deltas. Top-1 flips alone do not predict validation value: normalized allocation gain is 0.075 [0.004, 0.157] on flipped roots and 0.140 [0.049, 0.250] on stable roots. Thus response magnitude and decision relevance are not the same as allocation value.

The next question is whether available HF evidence can resolve the fragile decision. For each root, the ex-post decision-critical pair is the union of the proxy and deployment top-1 candidates, while the proxy top-2 margin measures the separation of the leading proxy candidates. At the primary long budget, the PIVOT query set intersects the decision-critical pair in 0.97 [0.92, 1.00] of roots, but the mean queried HF noise is 13.99 [7.97, 22.90] times the proxy top-2 margin. Thus targeting a decision-critical candidate does not by itself guarantee that the available HF evidence can resolve the ordering. Additional root-level diagnostics appear in Appendix B.

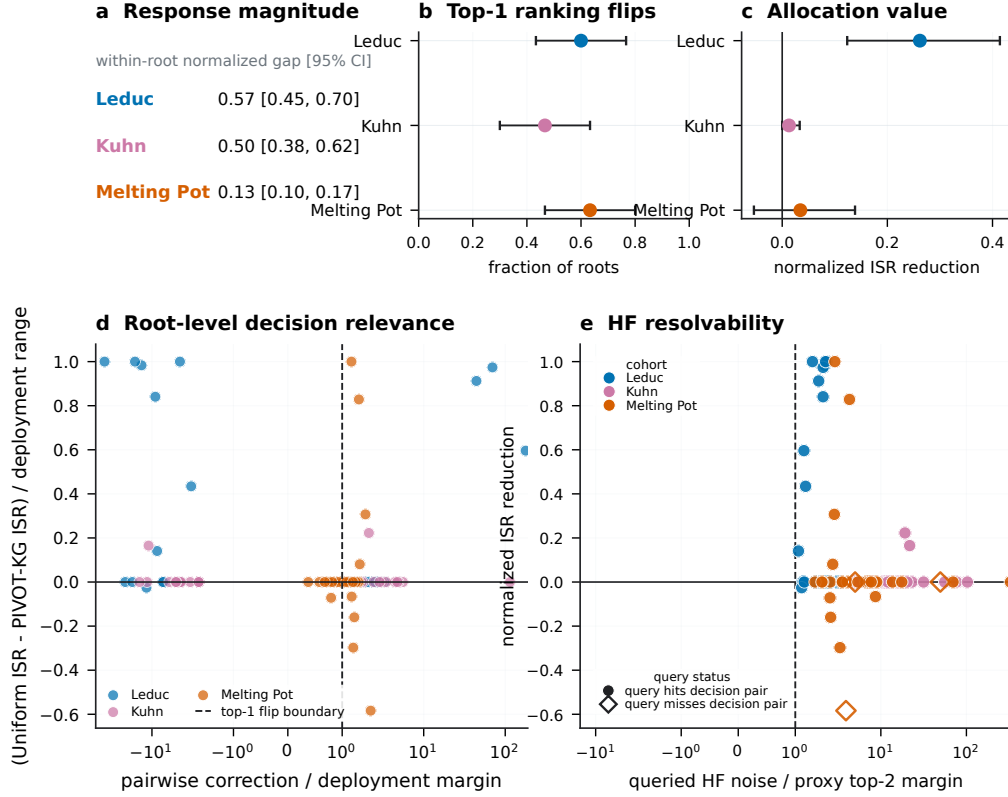


Figure 3: Response magnitude is not allocation value. Panel a reports within-root response gaps; b reports proxy/deployment top-1 flips; c reports normalized ISR reduction; d relates correction-to-margin ratio to normalized gain; e relates queried HF noise to the proxy top-2 margin. Filled circles denote queries intersecting the ex-post decision-critical pair and open diamonds denote misses.

Together, the sealed analysis shows that fidelity error creates validation value only when it threatens the update decision and the available HF evidence can resolve that uncertainty.

6.4 PIVOT-KG UNDER LIMITED HIGH-FIDELITY BUDGETS

Table 1 reports the sealed fixed-budget PIVOT-KG versus Uniform comparison. Leduc has a positive primary contrast, Kuhn is near-null, and the Melting Pot primary condition is unresolved with mixed secondary conditions. The pattern is consistent with the preceding mechanism analysis: a larger response gap does not by itself create allocation value.

Table 1: Sealed response-world allocation contrasts. Entries are paired PIVOT-KG minus Uniform gains with root-bootstrap 95% intervals; primary and secondary conditions are shown together.

World / condition	<i>n</i>	PIVOT-Uniform	95% CI	PIVOT ISR	Status
Leduc, long / 2 queries	30	0.036	[0.014, 0.064]	0.009	primary positive
Kuhn, long / 2 queries	30	0.002	[0.000, 0.006]	0.056	near-null
Melting Pot, long / 2 queries	30	0.906	[-0.784, 3.401]	0.836	primary null
Melting Pot, long / 1 query	30	3.423	[0.390, 6.944]	1.859	secondary positive
Melting Pot, short / 5 queries	30	-0.575	[-1.042, -0.180]	1.610	negative

For iterative self-improvement, we report cumulative improvement-selection regret, $CISR = \sum_{t=1}^T ISR_t$. Across 40 rounds, PIVOT-KG reduces paired CISR relative to Proxy Only by 1.4883 [1.0618, 1.9105]. Matched-cost comparisons with Global-VOI and Paired LUCB vary across settings; Figure 4 shows the resulting evidence-efficiency frontier together with an all-HF reference. Detailed closed-loop trajectories and robustness analyses appear in Appendix B.

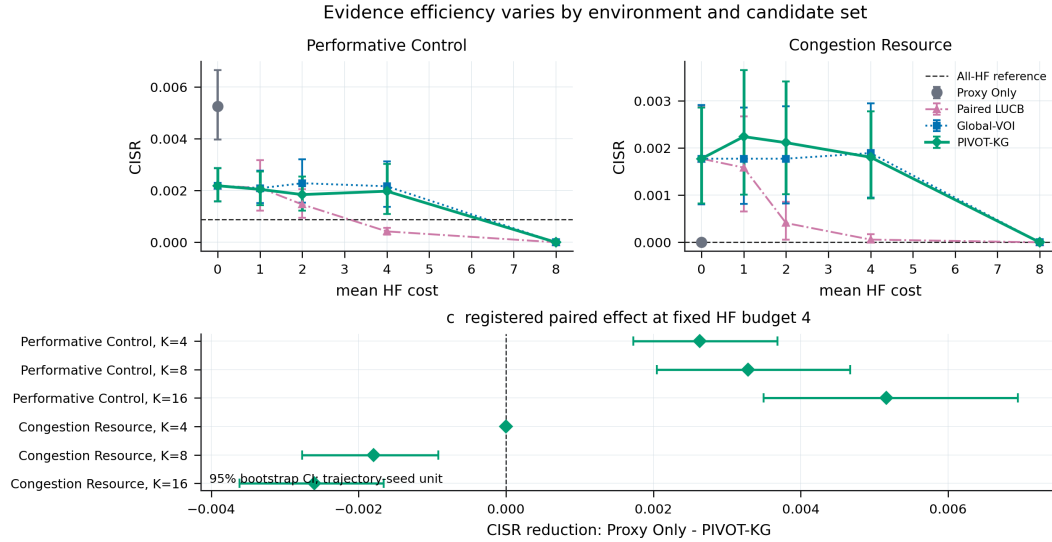


Figure 4: Evidence-efficiency frontier. Matched-cost curves and the fixed-budget forest report selection-regret reduction; the all-HF line is a reference. Intervals are seed bootstraps.

Overall, PIVOT-KG is useful when the available HF evidence can resolve uncertainty that matters for the update decision, rather than whenever deployment response is large.

6.5 EXTERNAL VALIDATION IN HIGHWAYENV

To test PIVOT-KG in a physically reactive setting where high-fidelity evaluation is genuinely scarce relative to the candidate set, we use a separately registered HighwayEnv redesign with eight distinct scheduled candidates, formed by crossing four non-idle actions with early and late half-horizon windows. The redesign fits the correction model on 40 calibration roots and evaluates 120 disjoint test roots. At the preregistered primary budget $B = 4$, PIVOT-KG reduces ISR relative to exact Uniform allocation by 0.0380 [0.0300, 0.0474]; the secondary $B = 2$ contrast is 0.0456 [0.0343, 0.0587]. Uniform is averaged exactly over all feasible queried subsets, and each PIVOT query corresponds to a physical paired actor rollout. Earlier five-action cohorts were unresolved at their registered primary budgets and are reported in Appendix C. We therefore treat the redesign as external evidence that targeted validation can help when candidate-set structure creates a meaningful allocation problem, not as a universal HighwayEnv advantage. Additional OOD-evaluator and public-data boundary analyses appear in Appendix D.

7 CONCLUSION

An agent can be correct about the wrong world. This paper argues that reliable self-improvement requires evaluating proposed policy replacements at the update level, under the response they induce. Improvement Fidelity formalizes whether proxy improvement preserves deployment improvement on the updates an agent actually proposes, and the analysis shows how operator shift, deployment response, and candidate margins determine when policy-level accuracy is insufficient. PIVOT-KG targets the resulting decision problem by allocating limited paired HF evidence according to its expected reduction in update-selection regret. Across controlled, sealed, and external settings, response magnitude alone does not determine either ranking failure or the value of additional validation.

Several boundaries remain. The direct, actor, and strategic response worlds cover selected forms of adaptation rather than all deployment dynamics, and the finite strategic fixtures are not equilibrium characterizations. PIVOT-KG also depends on the informativeness of HF evidence: limited budgets or noisy measurements can leave decision-relevant uncertainty unresolved. Finally, earlier five-action HighwayEnv cohorts are unresolved at their registered primary budgets, while the public-finance analysis is observational and does not identify a causal deployment response. The eight-candidate

HighwayEnv redesign has a positive primary $B = 4$ contrast in a panel deliberately large enough to create a nontrivial allocation problem, but its claim is limited to that registered panel and budget allocation.

These boundaries point to richer response models, repeated settings in which the response itself evolves with the agent, and intervention-based HF evidence in more realistic deployment systems. More broadly, self-improvement should be judged not only by whether a candidate scores better under its verifier, but by whether the proposed replacement remains better in the world it creates. Reliable self-improvement therefore requires fidelity of improvements, not merely accuracy of policy evaluations.

AI USE STATEMENT

The original version of this manuscript was written entirely by the authors. The research questions, conceptual framing, mathematical development, and experimental design were developed by the authors. Generative AI tools were used only during manuscript revision for language and presentation polishing; they were not used to formulate the research ideas, design the experiments, or generate numerical results. All reported experimental results were produced by the authors' executed research code and were checked and verified by the authors. The authors reviewed the final manuscript and take full responsibility for its contents.

Reproducibility Statement. The anonymous supplement contains deterministic configurations, hash-indexed source evidence, figure tables, and rebuild commands. The public-finance audit uses virtual fills and contains no credentials, private data, raw vendor archives, or live orders.

Ethics Statement. The study uses controlled virtual environments and public market data only; it involves no human subjects, personal data, live trading, or external authorization.

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APPENDIX

The appendix follows the main evaluation. Appendix A extends the response analyses from Section 6.2; Appendix B contains detailed PIVOT-KG and robustness results supporting Sections 6.3–6.4; Appendix C collects the complete HighwayEnv historical-cohort and redesign audit; Appendix D reports boundary analyses; and Appendix E summarizes statistical and reproducibility details.

A EXTENDED FIDELITY AND RESPONSE ANALYSES

The main text separates operator shift from policy-induced deployment response. Here we expand the response side of that analysis. The world-response decomposition traces matched observations through the direct, actor, and strategic levels, while the opponent-family analysis shows how the strategic effect varies across adaptive mechanisms. For the latter, the reported family-balanced strategic reversal rate is the unweighted mean across the three adaptive opponent families.

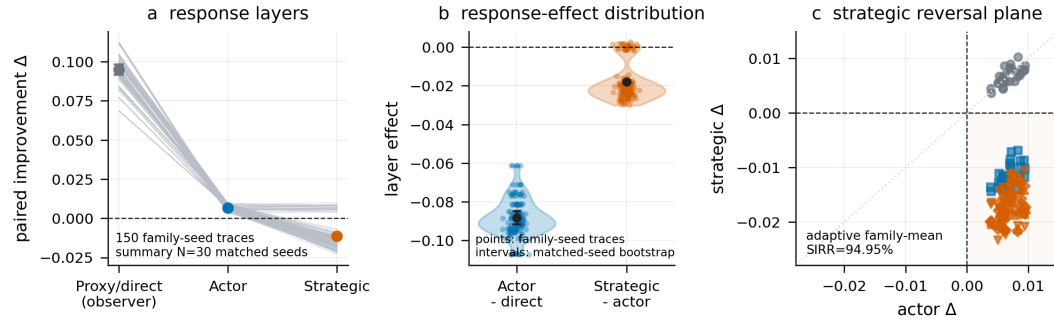


Figure 5: World-response decomposition. Panel a traces 150 family-by-seed observations (five opponent families $\times$ 30 matched seeds) through direct, actor, and strategic worlds; summary intervals use the 30 matched seeds. Panel b separates mechanical and strategic effects; panel c marks the strategic reversal quadrant.

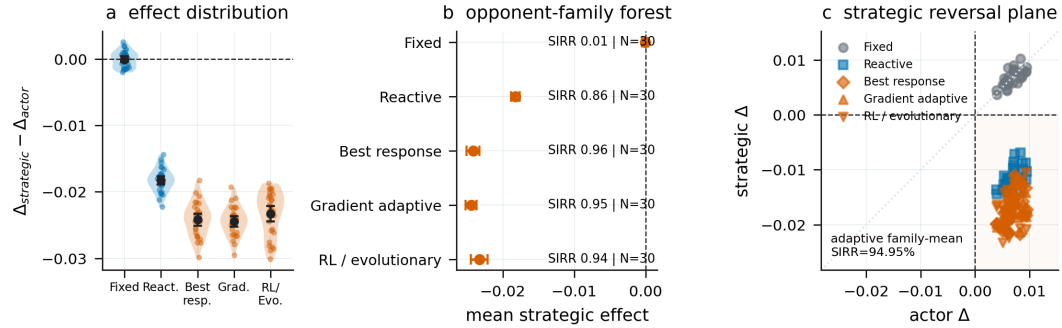


Figure 6: Strategic generalization across opponent mechanisms. Raincloud and forest views show family-specific effects and reversal rates over 30 matched opponent-seed clusters. The reversal plane labels the unweighted mean across the three adaptive families and uses both color and marker shape for grayscale legibility.

B EXTENDED PIVOT-KG EVALUATION

The main text reports the sealed allocation results and the matched-cost frontier. Here we provide additional decision-resolvability diagnostics, trajectory-level results, and numerical-robustness views. The within-root normalized response gaps are 0.57 [0.45, 0.70] for Leduc, 0.50 [0.38, 0.62] for Kuhn, and 0.13 [0.10, 0.17] for Melting Pot. At the primary long budget, the decision-critical query-hit rates are 1.00 in Leduc, 1.00 in Kuhn, and 0.90 in Melting Pot. The descriptive normalized gain is 0.113 [0.055, 0.180] among the 87 hit roots and -0.195 [-0.584, 0.000] among the three misses; the

final sealed estimate separation is 0.06 [0.05, 0.08] after root scaling. For the closed-loop plots, CTI denotes cumulative true improvement and CISR denotes cumulative improvement-selection regret.

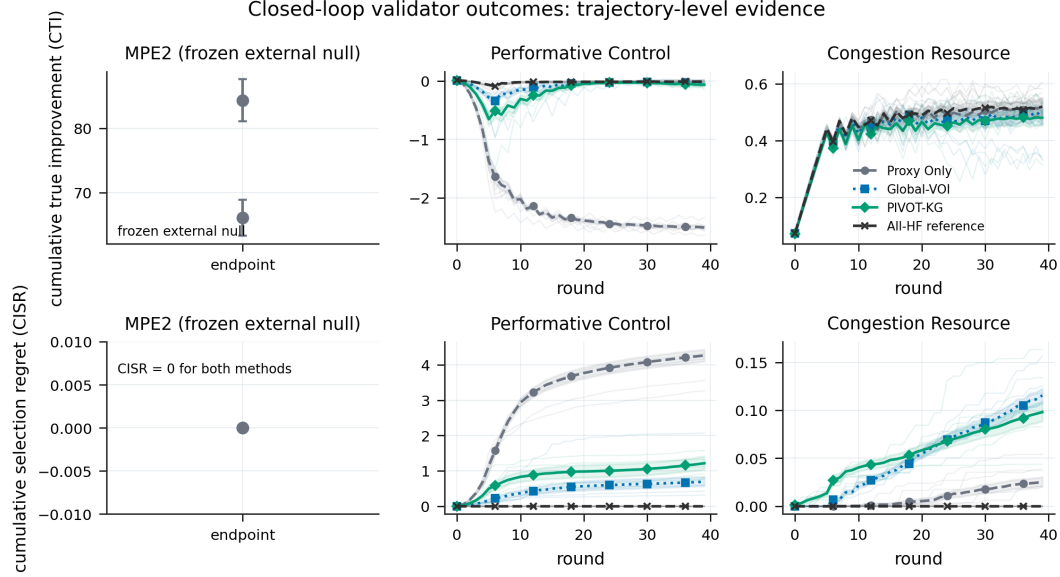


Figure 7: Closed-loop validator outcomes. The top row reports cumulative true improvement (CTI) and the bottom row cumulative improvement-selection regret (CISR); thin curves show seed-level trajectories and markers summarize the compared decision rules.

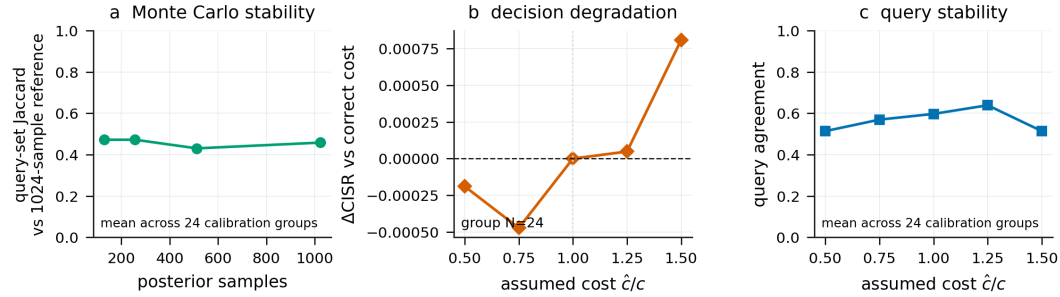


Figure 8: PIVOT-KG sensitivity to posterior sample count and cost misspecification across the registered calibration groups.

C HIGHWAYENV VALIDATION, REPLICATION, AND REDESIGN

The original and replication cohorts each contain 20 calibration and 60 test roots, with no overlap between cohorts. The replication freezes the five-action panel, four lanes, 20 traffic vehicles, horizon 20, and covariance shrinkage 0.5. It refits the correction prior on its own calibration cohort before opening the full test cohort. The primary contrast is Uniform minus PIVOT-KG ISR at $B = 2$; $B = 4$ is secondary. We retain every registered root and use 4,000 paired bootstrap resamples with fixed RNG seeds. Individual 95% intervals condition on each fitted calibration model and do not include calibration sampling uncertainty.

A technical smoke run exposed the first two replication test roots before the full run; those roots are retained without subsequent tuning or exclusions. The supplement contains the execution note, complete seed records, query and truth ledgers, dependency lock, frozen simulator code, and a standalone recomputation command. Calibration and post-decision truth audits are outside the logical query budget and are included in physical evaluation accounting. Uniform HF retains proxy estimates for unqueried candidates, whereas PIVOT-KG uses its calibrated posterior; the comparison therefore evaluates the complete selection rules rather than acquisition alone.

Eight-candidate redesign. The redesign freezes eight distinct scheduled candidates, crossing the four non-idle HighwayEnv actions with early and late half-horizon windows. It fits a joint correction model on 40 calibration roots and evaluates 120 disjoint test roots. The primary budget is $B = 4$; $B = 2$ is secondary. Uniform is the exact average over all $\binom{8}{B}$ queried subsets (70 at $B = 4$, 28 at $B = 2$), whereas PIVOT-KG uses sequential exact affine-normal Gaussian KG updates. Each selected query is a physical paired actor rollout. All candidate truth outcomes, including the shared incumbent, are evaluated only after the adaptive decisions and are excluded from the logical query budget. The audit checks all 3,840 physical evaluations and uses 4,000 paired bootstrap resamples with fixed seeds. Two test roots were exposed during technical smoke; they remain in the formal cohort without retuning or exclusion, so this is external DEV evidence rather than a wholly blinded confirmatory test. The supplement contains the hash-indexed evidence, frozen redesign source, query/truth ledgers, and independent table/figure audit.

Table 2: Original HighwayEnv study, fresh-seed replication, and the eight-candidate redesign. B counts candidate paired HF queries per decision. Positive contrasts favor PIVOT-KG; intervals are paired 95% bootstrap intervals. $B = 2$ is primary in the original/replication cohorts; $B = 4$ is primary in the redesign.

Cohort	HF budget	n	Uniform ISR	PIVOT ISR	Contrast	95% CI
Original	1	60	0.0492	0.0718	-0.0226	[-0.0439, -0.0008]
Original	2 (primary)	60	0.0422	0.0374	0.0049	[-0.0088, 0.0199]
Original	4	60	0.0144	0.0029	0.0115	[0.0008, 0.0237]
Replication	2 (primary)	60	0.0254	0.0261	-0.0006	[-0.0143, 0.0145]
Replication	4	60	0.0037	0.0004	0.0034	[-0.0007, 0.0099]
Redesign (8)	2 (secondary)	120	0.0613	0.0157	0.0456	[0.0343, 0.0587]
Redesign (8)	4 (primary)	120	0.0435	0.0055	0.0380	[0.0300, 0.0474]

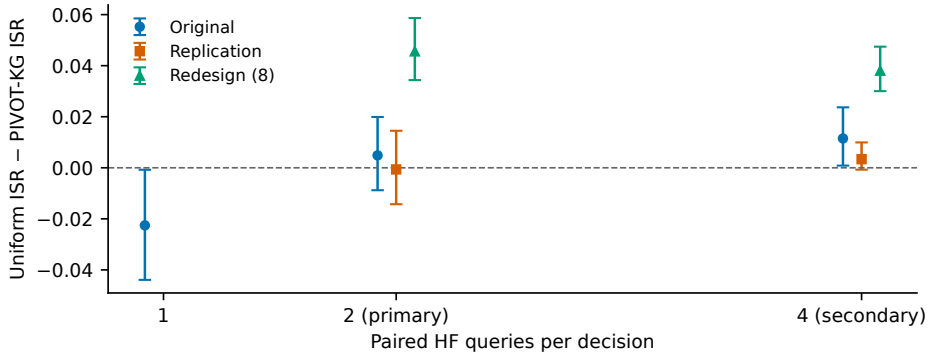


Figure 9: HighwayEnv paired allocation contrasts across the historical, fresh-seed, and eight-candidate redesign cohorts. Positive Uniform-minus-PIVOT ISR favors PIVOT-KG; error bars are individual 95% bootstrap intervals over the registered test seeds.

D BOUNDARY ANALYSES

OOD evaluator comparison. The powered OOD transition-evaluator comparison gives transition-minus-global ISC of -0.2590 [-0.3588, -0.1707]. This result separates the update-level estimand from any claim that a transition-specific learner must dominate a global learner.

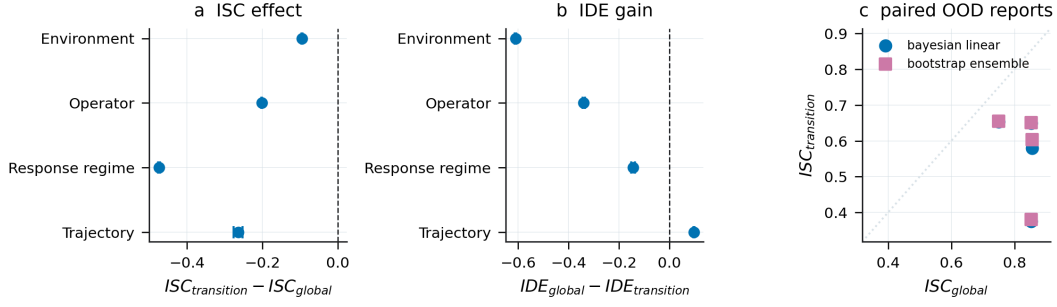


Figure 10: OOD evaluator comparison. Forest horizontal ranges are descriptive spans across the two fitted model families in each split; the scatter retains paired reports and the diagonal indicates equal ISC.

Public-finance diagnostic. The public-data audit uses historical paths, virtual fills, and a participation/depth proxy. It has 0/7 depth-proxy reversals in the expanded primary sessions and 0/5 in the holdout; the pooled depth-minus-replay effect is approximately -4.16×10^{-7} in native session units. Historical paths do not identify the counterfactual market response to deploying a different policy, so this analysis serves as an observational boundary check.

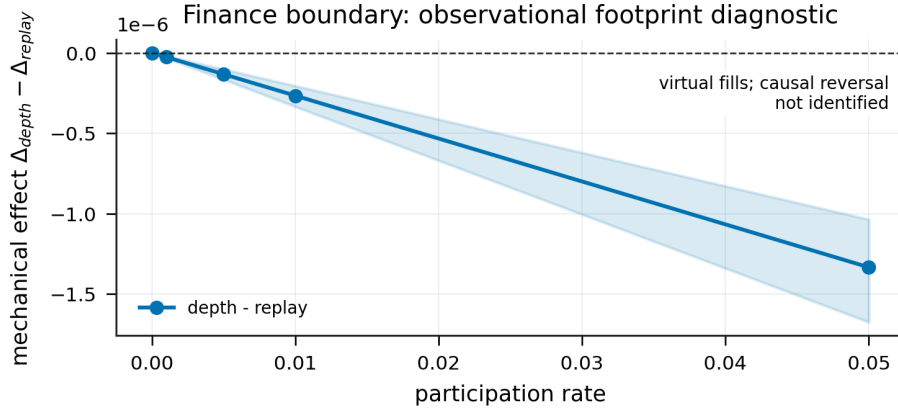


Figure 11: Initial single-asset public-finance boundary diagnostic over participation. Shading is a seven-session descriptive interval; fills are virtual. The separate 12-session, three-asset expansion is summarized in Table 3.

Table 3: Public-data boundary summary.

Public-data boundary	Value
Initial single-asset sessions	7
Expanded primary sessions	12
Expanded primary reversals / eligible	0/7
Expanded holdout reversals / eligible	0/5
Expanded pooled depth effect	-4.16e-07
Causal impact identified	No

E STATISTICAL AND REPRODUCIBILITY DETAILS

The release includes frozen source snapshots, result manifests, figure assets, source tables, and analysis scripts. Metadata record the unit of inference and distinguish descriptive spans from bootstrap intervals. The canonical metric implementation is the frozen statistics module. Closed-loop CTI/CISR values sum over rounds, whereas the budget frontier is a one-set ($T = 1$) selection problem, so their scales are reported separately. Internal experiment identifiers, configuration digests, exact source hashes, and raw sampled operator-shift cells are retained in the supplement and figure metadata for audit.

Table 4: Controlled evidence summary.

Evidence layer	Estimate	95% interval	Unit / interpretation
Operator shift	0.2627	[0.1684, 0.3763]	IDE contrast; paired transition
Closed-loop validation	1.4883	[1.0618, 1.9105]	CISR reduction; trajectory seed
OOD evaluator contrast	-0.2590	[-0.3588, -0.1707]	transition ISC minus global ISC; null
Strategic adaptation	-0.0240	[-0.0249, -0.0231]	strategic minus actor; opponent cluster